The Metric the Others Were Using

Information geometry: Fisher's metric and the dual connections of softmax

One could sum up this whole document in a sentence:

> The matrix that drove the training gradient, the second-order shape of the KL-divergence, and the one coordinate-invariant way to measure distance between distributions are all the same object — the Fisher information metric — and the space of softmax distributions carries not one but a dual *pair* of flat connections, one straight in logits and one straight in probabilities.

This is the fourth of a set. *Next-Token Prediction is Geometry* built the Euclidean picture (inner products, the softmax Jacobian, the spring-law gradient); *Training is a Monte Carlo Estimate* said what the loss measures (cross-entropy, KL); *The Radon–Nikodym Theorem* said what the densities are. Each used a geometry on the space of distributions without naming it. This note names it. The field is *information geometry* (Chentsov, Amari, Nagaoka).

Two manifolds, and which one carries the geometry

"The manifold" has meant two different things, and they carry different structure:

- the **representation manifold** — hidden states h and token embeddings e_w as points in $\mathbb{R}^d$, the "meaning manifold" of the first note, a curved subset of Euclidean space;
- the **statistical manifold** — the *output distributions* themselves, the family $\{\, q = \mathrm{softmax}(z) \,\}$ living in the probability simplex, each point a complete next-token distribution.

The canonical geometry lives on the statistical manifold. The representation manifold inherits a meaningful geometry only by *pulling back* from it — the last section. We do the statistical manifold first, because that is where every answer is clean.

The Fisher metric

Parametrise a family of distributions by θ and write the log-likelihood $\ell_\theta = \log q_\theta$. The *score* is its gradient $\partial_i \ell_\theta$, and the **Fisher information metric** is the covariance of the score:

$$g_{ij}(\theta) = \mathbb{E}_{q_\theta}\big[\partial_i \ell_\theta \, \partial_j \ell_\theta\big] = -\,\mathbb{E}_{q_\theta}\big[\partial_i \partial_j \ell_\theta\big].$$

It is symmetric and positive semi-definite — a Riemannian metric on the space of distributions. Three facts turn "a metric" into "*the* metric," and the second is the spine of this note.

It is the second-order shape of the KL-divergence

Expand the divergence between two nearby distributions. The first-order term vanishes (the divergence has a minimum of zero at coincidence), and the first surviving term is the Fisher

quadratic form:

$$D_{\mathrm{KL}}(q_\theta \,\|\, q_{\theta+\delta}) = \tfrac{1}{2}\,\delta^\top g(\theta)\,\delta + O(\|\delta\|^3).$$

The KL-divergence of the companion notes is not a metric globally — asymmetric, no triangle inequality — but infinitesimally it *is* one: g is precisely the thing being squared. Distance-between-distributions and the loss we descended are the same geometry at two scales.

For softmax it is exactly the Jacobian from the first note

Write the softmax family as the exponential family it is. With logits z as parameters,

$$\ell_z(w) = \log q_z(w) = z_w - A(z), \qquad A(z) = \log \textstyle\sum_k e^{z_k},$$

where A — the log-partition function — is the $\log\sum\exp$ we keep meeting. Its gradient is the distribution, $\partial_i A = p_i$, so the score is $\partial_i \ell_z(w) = \delta_{iw} - p_i$, and the Fisher metric is its covariance:

$$g_{ij}(z) = \mathbb{E}_w\big[(\delta_{iw} - p_i)(\delta_{jw} - p_j)\big] = p_i\,\delta_{ij} - p_i p_j = \frac{\partial p_i}{\partial z_j}.$$

$$g(z) \;=\; \mathrm{diag}(p) - p\,p^\top \;=\; \nabla^2 A(z) \;=\; \frac{\partial\, p}{\partial\, z} \;=\; \nabla^2_\delta\, D_{\mathrm{KL}}\big|_{\delta=0}.$$

The Fisher metric of the softmax family, the Hessian of $\log\sum\exp$, the softmax Jacobian $p_i(\delta_{ij} - p_j)$ that drove the spring law, and the second-order shape of KL are one matrix written four ways.

The identity is forced, not coincidental. Softmax is the gradient of the log-partition, $p = \nabla A$, so its Jacobian is the Hessian $\nabla^2 A$ — and the Hessian of a log-partition is by definition the covariance of the sufficient statistic, which is the Fisher metric. The matrix the first note computed as "the derivative of softmax" and used to push embeddings around was the metric tensor the entire time; training did differential geometry without declaring it.

More than carried along, the metric is *applied* on the backward pass. Differentiating a bare softmax gives $\partial L/\partial z = g\,(\partial L/\partial p)$, and since the logits z are the natural coordinates θ and the probabilities p the expectation coordinates η (next section), $g = \partial\eta/\partial\theta$ is the Jacobian *between the dual coordinate systems*. So the softmax backward pass *is* the Fisher metric at work: converting the gradient written in probability-coordinates into the gradient written in logit-coordinates — index lowering by g. The familiar fused simplification $\partial L/\partial z = p - y$ is exactly that metric multiplication, carried out in closed form.

Insight One caveat keeps the identity honest, because it is the standard place to go wrong. This is the Fisher metric of the *output distribution* in *logit* coordinates — an $n \times n$ matrix. It is *not* the parameter-space Fisher of natural-gradient fame, the (parameters × parameters) matrix $F_\theta = \mathbb{E}\big[\nabla_\theta \log q\ \nabla_\theta \log q^\top\big]$. That one is this matrix *pulled back* through the network,

$$F_\theta = J_{\mathrm{net}}^\top \big(\mathrm{diag}(p) - p\,p^\top\big)\, J_{\mathrm{net}}, \qquad J_{\mathrm{net}} = \partial z/\partial\theta,$$

the Gauss–Newton / K-FAC factorisation, and the pullback of §5. The bare softmax Jacobian is the *kernel* that gets sandwiched, not the sandwich.

Insight The metric is rank $n-1$, not n: adding a constant to every logit changes nothing, so g has the all-ones vector in its null space — the gauge freedom of softmax. On the tangent space of the simplex (perturbations whose entries sum to zero) it is strictly positive definite. The manifold is the $(n-1)$-dimensional simplex interior, and g is a genuine metric there; the degenerate direction is only the redundancy in the coordinates.

It is the only invariant metric (Chentsov)

Chentsov's theorem: on the simplex, the Fisher metric is the *unique* Riemannian metric, up to an overall scale, invariant under sufficient statistics — the maps that reparametrise without discarding information. If you want distance between distributions not to depend on arbitrary choices of coordinate, you do not get to choose: it is Fisher–Rao or nothing. This is what licenses the first note's whole premise that "geometry tracks meaning" — there is exactly one invariant geometry for it to track.

Dual coordinates: logits and probabilities are Legendre partners

An exponential family comes with two natural coordinate systems and a duality between them. The *natural* (canonical) parameters are the logits $\theta = z$; the *expectation* parameters are the probabilities themselves, $\eta = \mathbb{E}_w[\text{one-hot}] = p$. They are related by the gradient of the log-partition and its convex conjugate:

$$\eta = \nabla A(\theta) = \operatorname{softmax}(\theta), \qquad A^*(\eta) = \sup_\theta \big(\langle \theta, \eta\rangle - A(\theta)\big) = \sum_w p_w \log p_w = -H(p).$$

The dual potential is the *negative entropy* — the same H from the coding story. So the Legendre pair underneath softmax is

$$A = \log \sum \exp \quad \longleftrightarrow \quad A^* = -H,$$

and the Fisher metric is the Hessian of *either* potential, in its own coordinates: $g = \nabla^2 A(\theta)$ in logit-coordinates, $g^{-1} = \nabla^2 A^*(\eta)$ in probability-coordinates. The maximum-entropy characterisation of softmax (why $\nabla A = \operatorname{softmax}$) and the metric are two readings of one convex duality.

The connections: not one, but a dual pair

Here the statistical manifold is richer than an ordinary Riemannian one. A metric alone gives the Levi–Civita connection — one notion of "straight line" and parallel transport. An exponential family carries a whole one-parameter family of α-connections, and two are distinguished:

- the **exponential connection** (e-connection, $\alpha = +1$): its Christoffel symbols *vanish in the logit coordinates* θ. The family is e-flat, and straight lines *in logit space* are its geodesics.
- the **mixture connection** (m-connection, $\alpha = -1$): its Christoffel symbols *vanish in the probability coordinates* η. The family is m-flat, and straight lines *in probability space* are its geodesics.

These two are *dual* with respect to the Fisher metric — $\partial_k g_{ij} = \Gamma^e_{ki,j} + \Gamma^m_{kj,i}$ — and the Levi–Civita connection is exactly their average, the $\alpha = 0$ case. The softmax family is therefore **dually flat**: it has two distinct canonical notions of straightness, and the metric geodesic is neither, but their mean.

Insight Dual flatness is what being an exponential family *buys* geometrically, and it has teeth. In a dually flat manifold the KL-divergence is the **Bregman divergence** of the potential — $D_{\mathrm{KL}}(p\|q)$ is the gap between A^* at p and its tangent plane at q — and a **generalised Pythagorean theorem** holds:

$$D_{\mathrm{KL}}(P\|R) = D_{\mathrm{KL}}(P\|Q) + D_{\mathrm{KL}}(Q\|R)$$

whenever the m-geodesic from P to Q meets the e-geodesic from Q to R at a Fisher-orthogonal angle. That right angle is the engine under the EM algorithm, information projection, and variational inference: their decompositions are literal orthogonal projections in this metric. Parallel transport, too, splits in two — e-transport carries logit-vectors, m-transport carries probability-vectors, and they are adjoint under g.

The pullback: a metric on the representation manifold

Now the $\mathbb{R}^d$ manifold of hidden states. It trivially inherits the *Euclidean* metric of the ambient space — that is what the inner product $\langle h, e_w\rangle$ uses — but that metric is not canonical: it depends on the coordinates the network happened to learn and respects no invariance. The meaningful metric arrives by *pulling back* the Fisher metric through the map that turns a hidden state into a distribution. With logits $z = Wh$ (the unembedding), the pulled-back metric on hidden-state space is

$$\hat{g}(h) = W^\top\big(\mathrm{diag}(p) - p\,p^\top\big)W, \qquad p = \mathrm{softmax}(Wh),$$

which is the parameter-space Fisher F_θ of §2's caveat, restricted to the unembedding layer — the bare Jacobian $\mathrm{diag}(p)-pp^\top$ sandwiched by W. It has a direct reading through the KL expansion of §2:

$$D_{\mathrm{KL}}\big(q(h)\,\|\,q(h+dh)\big) = \tfrac{1}{2}\,dh^\top \hat{g}(h)\,dh + \cdots .$$

Two hidden states are near in $\hat{g}$ exactly when they induce *similar next-token distributions*. This is the distributional hypothesis — *you shall know a word by the company it keeps* — promoted from slogan to Riemannian metric: semantic distance is difference in predictive behaviour. And it is in general *not* the Euclidean distance on h; it is the Euclidean distance warped by how sharply the output reacts, the warp factor being the Fisher metric at p.

Insight This is the precise sense in which the first note's Euclidean picture was a chart, not the territory. The gradient $\partial L/\partial h = \bar{e} - e_y$ it derived is the *Euclidean* gradient; the geometry-respecting update is the *natural gradient* $\hat{g}^{-1}\,\partial L/\partial h$, which moves along the manifold rather than across the coordinates. Natural-gradient descent is ordinary gradient descent performed in the Fisher metric — doing the geometry the first note only gestured at.

Two honest caveats

The elegance should not oversell itself.

Flat ambient, curved slice. The full simplex is dually flat, but the submanifold the network actually sweeps out — the image of $h \mapsto \mathrm{softmax}(Wh)$ — is in general a *curved* submanifold of it. Its embedding curvature measures how far the model's own family is from being an

exponential family in its own right. "Flat" describes the ambient statistical manifold, not the model's slice through it.

Smoothness, and the density underneath. All of this assumes a smooth manifold. The first note already flagged that the "meaning manifold" is loose — ReLU kinks, varying rank, stratification — so the apparatus is airtight on the statistical manifold and only as rigorous as the map $h \mapsto q$ is nice on the representation side. And the Fisher metric is defined through $\ell = \log q$ and its score $\partial \log q$ — densities, which is to say Radon–Nikodym derivatives. In the finite, strictly-positive softmax world those exist for free and the metric is elementary; let the vocabulary become a continuum and whether the metric exists at all becomes the same Radon–Nikodym question the third note ends on. The geometry, like everything else in this set, rests on a foundation that is silent here and load-bearing the moment the sum becomes an integral.